

CORONAL MASS EJECTION is a bubble of plasma ejected from the Sun into space

CORONA is hundreds of times hotter than the photosphere

GRANULATION is the mottling of the surface caused by convection cells

FACULAE are intensely bright active regions that are associated with the appearance of sunspots

PROMINENCE is a dense cloud of gas, suspended above the Sun's surface by magnetic field loops, that may persist for days or even weeks

SPICULES are short-lived jets of gas that are 6,000 miles (10,000 km) long

SURFACE

The visible surface of the Sun is called the photosphere. It is a layer of plasma (ionized gas) about 60 miles (100 km) thick and appears granulated or bubbly. The bumps, which are about 600 miles (1,000 km) wide, are the upper surfaces of convection cells that bring hot plasma up from the Sun's interior. Other significant features of the photosphere are sunspots, which are cooler regions that appear dark against their brighter, hotter surroundings.

Sunspots and related phenomena, such as solar flares (tremendous explosions on the Sun's surface) and plasma loops, are thought to have a common underlying cause: they are associated with strong magnetic fields or disturbances in these fields. The magnetic fields result from the fact that the Sun is a rotating body that consists largely of electrically charged particles (ions in its plasma). Different parts of the Sun's convective zone rotate at different rates (faster at the equator than the poles), causing the magnetic field lines to become twisted and entangled over time. Sunspots are caused by concentrations of magnetic field lines inhibiting the flow of heat from the interior where they intersect the photosphere. Other types of disturbance are caused by twisted field lines popping out of the Sun's surface, releasing tremendous energy, or by plasma erupting as loops along magnetic field lines. The amount of sunspot and related activity varies from a minimum to a maximum over an 11-year cycle.

SUNSPOTS

Each sunspot has a dark central region, the umbra, and a lighter periphery, the penumbra. Away from the sunspots, the Sun's surface looks granulated. Each granule is the top of a convection cell in the Sun's interior.

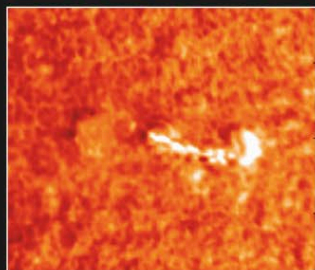
PHOTOSPHERE

The base of the photosphere has a temperature of 10,300°F (5,700°C), but its upper layers are cooler and emit less light. Here, the edge of the Sun's disk looks darker because light from it has emanated from these cooler regions.

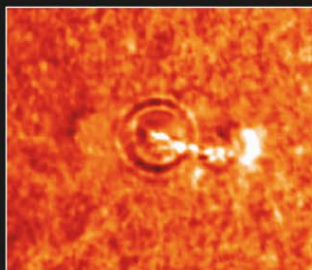
SOLAR ACTIVITY

This ultraviolet image of the Sun was obtained by an instrument onboard the SOHO solar observatory. It shows the Sun's chromosphere (the layer just above the photosphere) and various protuberances, including a huge solar prominence, as well as a number of active regions on the solar surface. The image also shows a coronal mass ejection with a bright central area of ultraviolet emission.

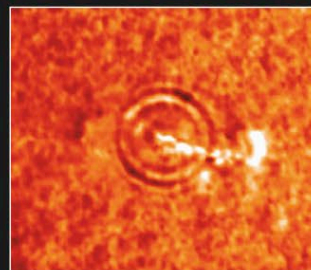
FIRST OBSERVATION OF A SOLAR QUAKE



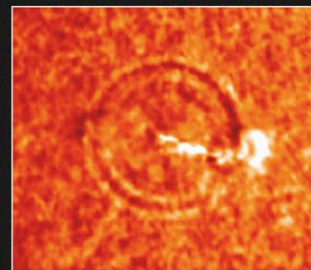
1 In July 1996, by analyzing data obtained by an instrument on the SOHO observatory, scientists recorded a solar quake for the first time.



2 The quake, equivalent to an earthquake of magnitude 11, was caused by a solar flare, visible as the white blob with a "tail" to its left.



3 The seismic waves looked like ripples on a pond but were 2 miles (3 km) high and reached a maximum speed of 248,600 mph (400,000 kph).



4 Over the course of an hour, the waves traveled a distance equal to 10 Earth diameters before fading into the fiery background.